5. These bacteria can be grown in a media containing specific antibiotics as per the resistant gene present in the plasmid vector. The bacterial cells which have taken up the recombinant plasmid vector can only grow in the presence of antibiotics (positive clones).
6. These positive clones can then be subjected to further applications like production of recombinant proteins, studying expression etc.,
7. This method has been successfully used for commercial production of many proteins like Human Insulin, antibodies, enzymes of medical and industrial interests etc.,

## 6.6 GENOME SEQUENCING AND THE BIRTH OF GENOMICS ERA

The birth of the 21$^{st}$ century has seen many revolutions in the field of molecular biology. Today molecular biologists are more interested in studying the collective behaviour of the genome rather than studying one individual gene. The entire information present in the DNA of an organism can be considered as its genome. Genomics can be defined as a study of whole genomes of organisms and uses a combination of techniques such as recombinant DNA technology, PCR amplification, Advanced DNA sequencing methods and bioinformatics analysis. Genomics experiments tries to harness the availability of entire sequence of genome which is possible today by the pioneering sequencing experiment by Fredrick Sanger and the recent next generation sequencing technologies most of which are build upon Sanger sequencing procedure

### 6.6.1 The Sanger sequencing procedure

1. It is also known as dideoxy chain termination method.
2. It sequence is generated by replicating a template DNA strand (the region to be sequenced) and interrupting the replication process at one of the four bases.
3. Four different reaction mixtures are produced such that the growing chain terminates in A,T, G or C
4. The reaction mixture also contains dideoxynucleoside triphosphates (ddNTPs) along with usual dNTPs.
5. If during replication ddNTPs is incorporated instead of usual dNTPs in the growing DNA strand then the replication stops at that nucleotide.
6. The ddNTPs are radio labelled (with $P_{32}$)or fluorescently labelled so the chain can be visualized in a gel electrophoresis procedure.

The details of Sanger sequencing procedure is depicted in figure 19 This sequencing procedure was performed in a massively parallel fashion and the human genome was sequenced in the year 2000.



XII U6 Mol Bio.indd 157 4/8/2020 3:16:27 PM